DNA associated with a core nucleosome, together with the spacer region, is approximately 200 bp.

In addition to the major histone types, a number of specialized histones can substitute for H3 or H2A and confer specific characteristics on the genomic DNA at that location. Histones can also be modified by chemical changes, and these modifications can change the properties of nucleosomes that contain them. As discussed further in Chapter 3, the pattern of major and specialized histone types and their modifications can vary from cell type to cell type and is thought to specify how DNA is packaged and how accessible it is to regulatory molecules that determine gene expression or other genome functions.

During the cell cycle, as we will see later in this chapter, chromosomes pass through orderly stages of condensation and decondensation. However, even when chromosomes are in their most decondensed state, in a stage of the cell cycle called **interphase,** DNA packaged in chromatin is substantially more condensed than it would be as a native, protein-free, double helix. Further, the long strings of nucleosomes are themselves compacted into a secondary helical structure, a cylindrical "solenoid" fiber (from the Greek *solenoeides*, pipe-shaped) that appears to be the fundamental unit of chromatin organization (see Fig. 2-5). The solenoids are themselves packed into **loops** or domains attached at intervals of approximately 100,000 bp (equivalent to 100 kilobase pairs [kb], because 1 kb = 1000 bp) to a protein **scaffold** within the nucleus. It has been speculated that these loops are the functional units of the genome and that the attachment points of each loop are specified along the chromosomal DNA. As we shall see, one level of control of gene expression depends on how DNA and genes are packaged into chromosomes and on their association with chromatin proteins in the packaging process.

The enormous amount of genomic DNA packaged into a chromosome can be appreciated when chromosomes are treated to release the DNA from the underlying protein scaffold (see Fig. 2-1). When DNA is released in this manner, long loops of DNA can be visualized, and the residual scaffolding can be seen to reproduce the outline of a typical chromosome.

The Mitochondrial Chromosome

As mentioned earlier, a small but important subset of genes encoded in the human genome resides in the cytoplasm in the mitochondria (see Fig. 2-1). Mitochondrial genes exhibit exclusively maternal inheritance (see Chapter 7). Human cells can have hundreds to thousands of mitochondria, each containing a number of copies of a small circular molecule, the mitochondrial chromosome. The mitochondrial DNA molecule is only 16 kb in length (just a tiny fraction of the length of even the smallest nuclear chromosome) and encodes only 37 genes. The products of these genes function in mitochondria, although the vast majority of proteins within the mitochondria are, in fact, the products of nuclear genes. Mutations in mitochondrial genes have been demonstrated in several maternally inherited as well as sporadic disorders (Case 33) (see Chapters 7 and 12).

The Human Genome Sequence

With a general understanding of the structure and clinical importance of chromosomes and the genes they carry, scientists turned attention to the identification of specific genes and their location in the human genome. From this broad effort emerged the **Human Genome Project,** an international consortium of hundreds of laboratories around the world, formed to determine and assemble the sequence of the 3.3 billion base pairs of DNA located among the 24 types of human chromosome.

Over the course of a decade and a half, powered by major developments in DNA-sequencing technology, large sequencing centers collaborated to assemble sequences of each chromosome. The genomes actually being sequenced came from several different individuals, and the consensus sequence that resulted at the conclusion of the Human Genome Project was reported in 2003 as a "reference" sequence assembly, to be used as a basis for later comparison with sequences of individual genomes. This reference sequence is maintained in publicly accessible databases to facilitate scientific discovery and its translation into useful advances for medicine. Genome sequences are typically presented in a 5′ to 3′ direction on just one of the two strands of the double helix, because—owing to the complementary nature of DNA structure described earlier—if one knows the sequence of one strand, one can infer the sequence of the other strand (Fig. 2-6).

Organization of the Human Genome

Chromosomes are not just a random collection of different types of genes and other DNA sequences. Regions of the genome with similar characteristics tend to be clustered together, and the functional organization of the genome reflects its structural organization and sequence. Some chromosome regions, or even whole chromosomes, are high in gene content ("gene rich"), whereas others are low ("gene poor") (Fig. 2-7). The clinical consequences of abnormalities of genome structure reflect the specific nature of the genes and sequences involved. Thus abnormalities of gene-rich chromosomes or chromosomal regions tend to be much more severe clinically than similar-sized defects involving gene-poor parts of the genome.

As a result of knowledge gained from the Human Genome Project, it is apparent that the organization of DNA in the human genome is both more varied and